25. Quantum query complexity with explicit oracle and output models

Pure Thought research portfolio

Revised 17 September 2026

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Revised specification — 2026-09-11. Detailed critique · Original description · Evidence standard

Plain-language guide

The problem in everyday terms

A claim that a quantum computer is faster is incomplete until we specify what information it receives, what answer it must return, and which operations we count. Here the problem is deliberately simple: find one marked item in a list when the only way to learn about an item is to query it. We compare a quantum search procedure with a proof of how many queries any procedure of the chosen kind must use.

Key terms

- **Oracle** — A precisely defined black-box operation providing access to input information; it is a mathematical access model, not a prediction device.
- **Query** — One use of the oracle, counted separately from other computation.
- **Promise problem** — A problem whose inputs are guaranteed to satisfy a stated restriction, here exactly one marked item.
- **Grover search** — A quantum algorithm that amplifies the probability of observing a marked item using repeated oracle and interference operations.
- **Amplitude** — A quantum coefficient whose squared magnitude contributes to an outcome probability.
- **Query complexity** — The number of oracle uses required to solve a problem with the specified success probability.
- **Upper and lower complexity bounds** — An algorithm showing that a cost suffices, and a proof that no allowed algorithm can beat a stated cost.
- **BPP and BQP** — Classes of problems efficiently solvable with bounded error by classical randomized and quantum computation, respectively.
- **Oracle separation** — A proof that two classes differ when both have access to a particular oracle; it does not by itself separate the ordinary classes.
- **Gate complexity** — The number of elementary circuit operations, a different resource from oracle queries.

Why this matters

This is a clean setting in which a quantum advantage can be proved rather than inferred from a small simulation. It also teaches how to assess broader speedup claims fairly. Input preparation, precision and output requirements can dominate a real application even when a query bound looks attractive.

What progress would mean

Success would combine a working small implementation with an all-size argument matching upper and lower query bounds. That establishes an advantage in this access model, not a blanket claim that quantum computers speed up every search task.

Core question

For unstructured search with a specified promise, can an implementation and an analytic lower bound establish a matched quantum query complexity?

Scope and assumptions

Use N=2^n items with exactly one marked item and phase-oracle access O_f|x⟩=(−1)^f(x)|x⟩. State success probability at least 2/3 and count oracle uses separately from elementary gates, state preparation, and output verification. Begin with small exact simulations and a proof valid for all N.

Retain quantum walks, linear systems, QAOA and oracle separations as independent extensions, each with its own input/output and cost model. Classical simulation wall time is not quantum runtime.

Mathematical target

Let θ=arcsin($N^{-1/2}$). After r Grover iterations the success probability is $\sin^2((2r+1)\theta)$. Choose r as a nearest nonnegative integer to $\pi/(4\theta)-1/2$, evaluating neighboring integers and handling small N explicitly. Derive the query upper bound and pair it with a hybrid/adversary/polynomial lower bound $\Omega(\sqrt{N})$ under the same promise.

An oracle separation must be written $BQP^{O}\neq BPP^{O}$ for a specified O; it does not establish BQP≠BPP. A linear-system extension must state matrix access, κ, precision, state preparation, and whether the output is a quantum state or classical vector.

Required outputs and proof obligations

Return oracle/circuit specification, exact small-instance results, an all-N success derivation, query/gate/resource accounting, and the lower-bound proof with assumptions. Any QAOA extension must distinguish a certified bound on a finite graph from a performance theorem for a graph family.

Validation and rejection controls

Check the analytic success formula on n=1,…,8 and all marked positions using symmetry or exhaustive checks. Test neighboring iteration counts to expose overshoot. Verify unitarity and normalization. Compare with randomized classical search under identical oracle access and error tolerance.

Milestones and research extension

Baseline: Grover construction and matched query bound. Strong: one second promise problem with matched assumptions and a certified upper/lower bracket. Research extension: a new restricted

query bound or oracle result. Reimplementation of established algorithms is replication, not a new complexity separation.

Milestones are acceptance gates, not calendar promises. Before a run, freeze the concrete instance/domain, tolerance, compute budget, and permitted dependencies. A bounded search that fails to find a result must preserve an unresolved outcome.

Starting instruction

Work on the baseline above. First make every input and convention explicit, then validate the reference case. Build the checker alongside the calculation. Label numerical evidence, finite-model certificates, and continuum theorems separately; return the strongest justified result and identify the exact unmet proof obligation.

Primary references and their role

- Bennett et al., Strengths and Weaknesses of Quantum Computing — Oracle lower-bound methodology for quantum search.

References support definitions or limitations; the scope choices and proposed milestones are editorial recommendations, not claims that these results have already been obtained.

Displayed formulation and claim boundary

Canonical specification.

$$\theta = \arcsin(N^{-1/2}), \qquad p_r = \sin^2((2r+1)\theta), \qquad r \approx \frac{\pi}{4\theta} - \frac{1}{2}.$$

Choose a nonnegative integer by checking neighboring counts and explicitly handle small instances under the declared oracle model. The target is a matched $O(\sqrt{N})$ algorithm and $\Omega(\sqrt{N})$ lower bound with identical promise and error tolerance. An oracle separation has the form $\mathrm{BQP}^O \neq \mathrm{BPP}^O$ and does not prove BQP $\neq$ BPP.

Detailed review and rationale

Revised problem · Preserved original · Pinned upstream source

Line references below refer to the preserved original at commit `f854d4e12b55`, not the rewritten file. "Blocking" means the original issue can invalidate the scientific target or claimed certificate; "major" means it materially changes scope, inference, or acceptance criteria. These are critiques of specifications, not results of running the embedded research-code sketches.

Detailed findings

1. **Blocking — unrelativized separation (25, 41).** An oracle separation does not prove BQP≠BPP. The superscript and access model belong in every theorem statement.
2. **Major — mixed objectives (23–27).** Grover, HHL, walks and QAOA solve different input/output problems; a single "quantum advantage" benchmark cannot fairly compare them without cost models.

3. **Major — Grover rounding (79–81).** The optimal iteration rule must account for the $-1/2$ shift and nearest-integer choice; the stated floor can overshoot. The exact success formula is the acceptance test.
4. **Major — HHL/QAOA scaling (16, 1035, 1052).** Logarithmic dimension dependence alone omits input access, κ, error and readout costs. A finite sample of QAOA graphs does not establish an approximation guarantee. The rewritten core provides one well-posed theorem target and modular extensions.

What the revision preserves and changes

The central motivation is retained, but the executable question is now: **For unstructured search with a specified promise, can an implementation and an analytic lower bound establish a matched quantum query complexity?** The revised scope chooses a concrete starting model; those choices are proposed research-design decisions, not facts inferred from the original. The baseline can succeed without resolving the full open research program.

The replacement also removes the original implementation sketches from the active specification. They remain in the archive for comparison, but have not been made executable or certified. New implementations should derive their equations from the revised definitions rather than copy unchecked prototypes.

Acceptance criteria to use instead

Baseline: Grover construction and matched query bound. Strong: one second promise problem with matched assumptions and a certified upper/lower bracket. Research extension: a new restricted query bound or oracle result. Reimplementation of established algorithms is replication, not a new complexity separation.

Sources supporting the corrections

- Bennett et al., Strengths and Weaknesses of Quantum Computing — Oracle lower-bound methodology for quantum search.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented

trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means “not excluded”; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.

- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.